Geographic Dispersion of Economic Shocks: Evidence from the Fracking Revolution

By Feyrer, Mansur, Sacerdote (AER 2017)

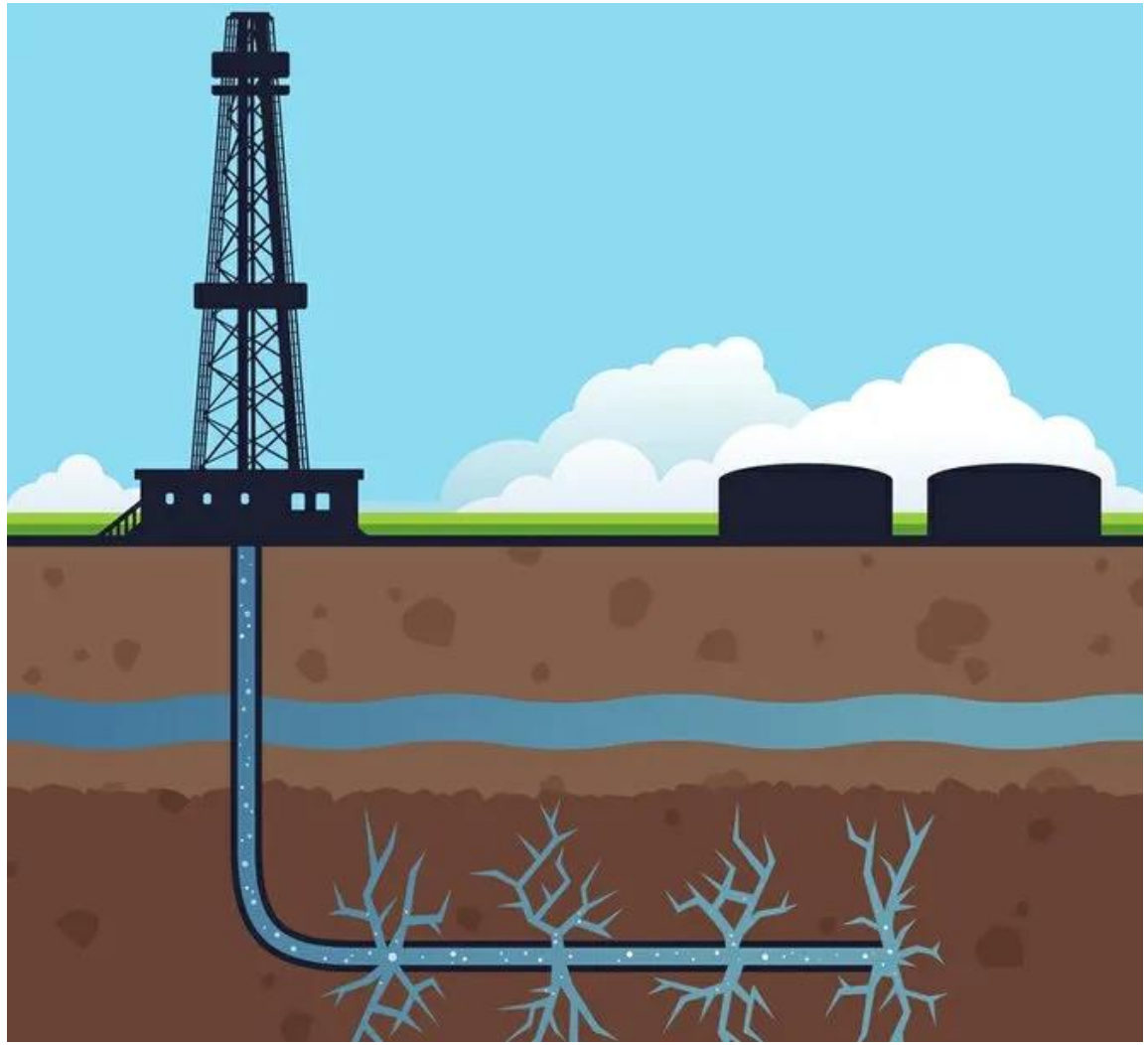
Quan Nguyen

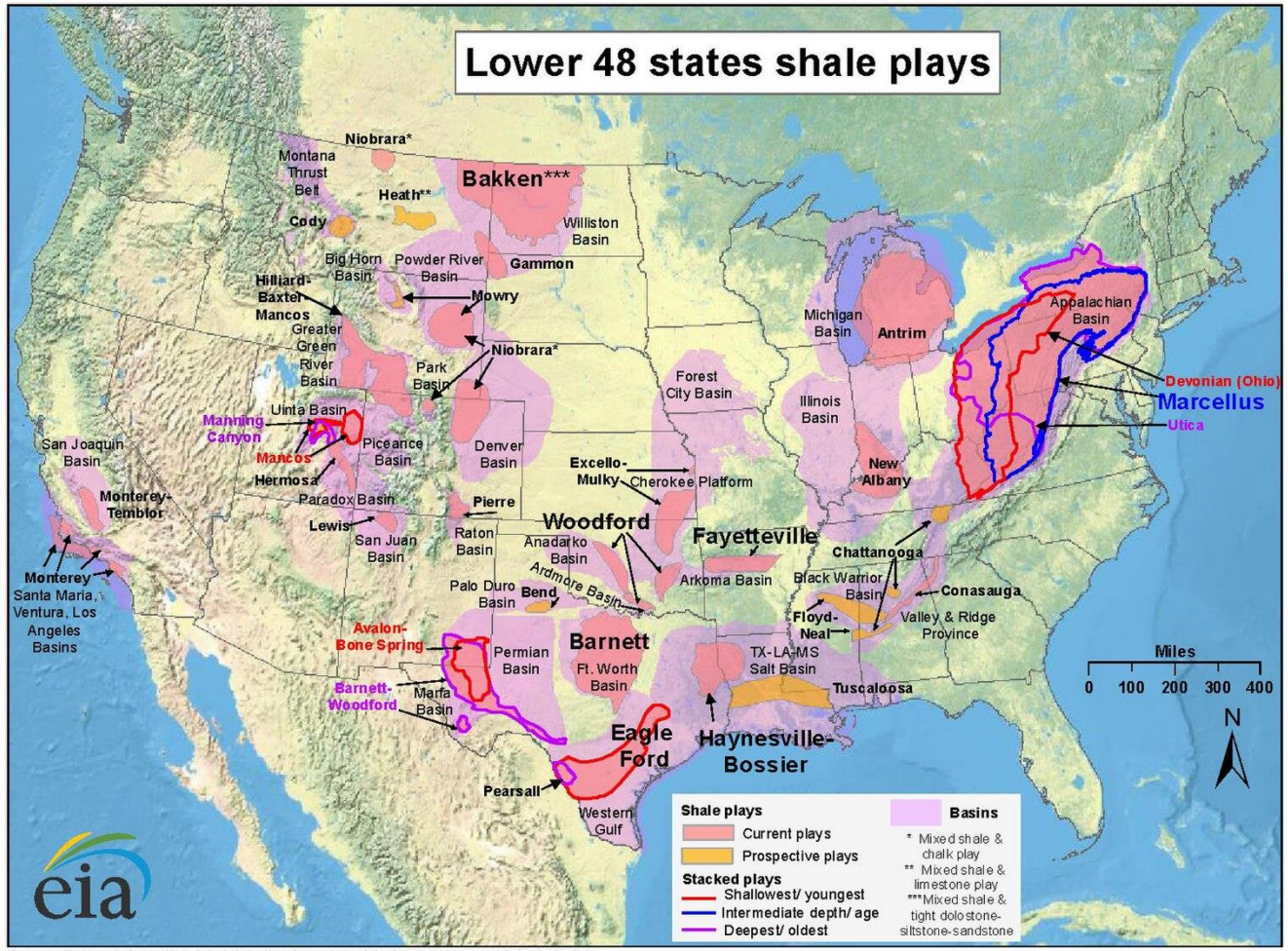
Research Question

- Using fracking as a natural experiment, we ask: when new oil and gas production occurs, **how much of the economic value stays local, who benefits, how far do the effects travel across space and time?**
- Knowing what happens to these dimensions gives operational guidance to policy makers.
- Overall contributes to the literature on the economic responses to local shocks.

Fracking/Shale Revolution

- A bundle of technologies took off in the mid 2000s - early 2010s
  - Horizontal drilling
  - 3-D seismic imaging
  - **Hydraulic fracturing**
- Natural gas/oil rich shale formations that were hard to extract before became extractable
- Resulted in massive sudden production growth → the U.S. becoming a dominant energy producer





Source: Energy Information Administration based on data from various published studies.
Updated: May 9, 2011

- **Shale Play:** is a geological area within a petroleum basin where oil and natural gas resources are trapped within a low-permeability shale rock formation
- Timing of plays taking off differed

Data

Data

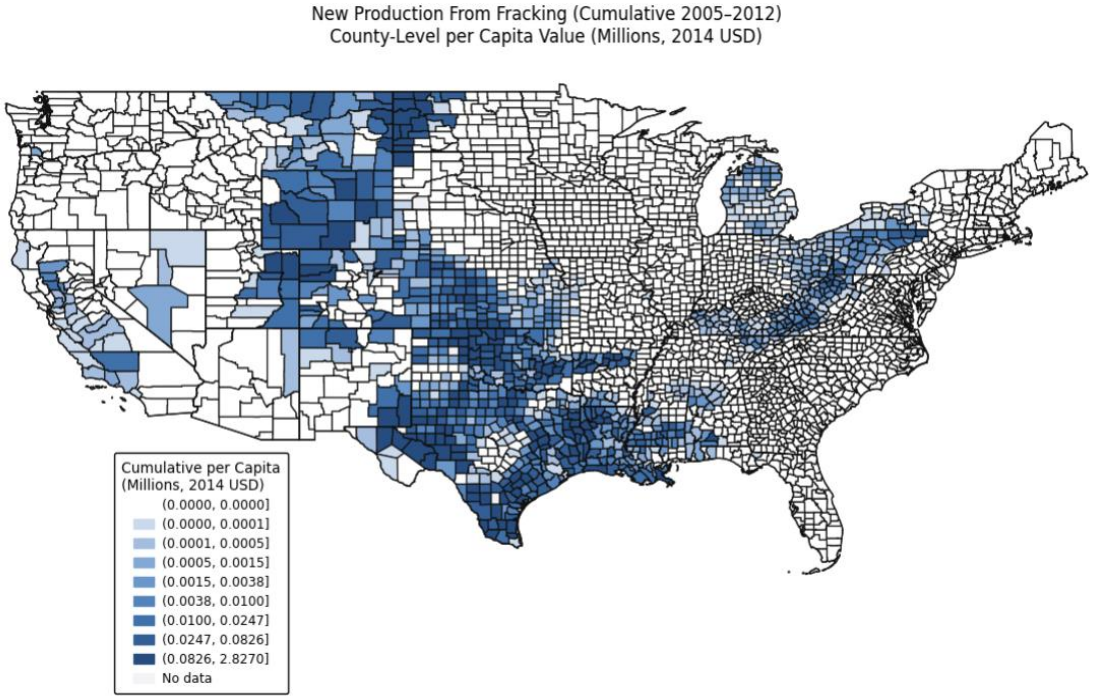
<u>Data Sources:</u>

- Drilling Data (Production)
  - *Source:* DrillingInfo well-level data (oil/gas by well × month × county)
- Labor Market Data (Wages & Employment)
  - *Source:* BLS Quarterly Census of Employment & Wages
- Household Income Data (Royalties & Business Income)
  - *Source:* IRS Statistics of Income

<u>Notes on sample construction:</u>

- Focus on 2005–2012 when all data sources overlap.
- Exclude smallest 2% of counties (very low employment)
- Scale all variables by lagged county employment for comparability across counties of different sizes.

Only 25% of counties from 2005-2012 saw any new production



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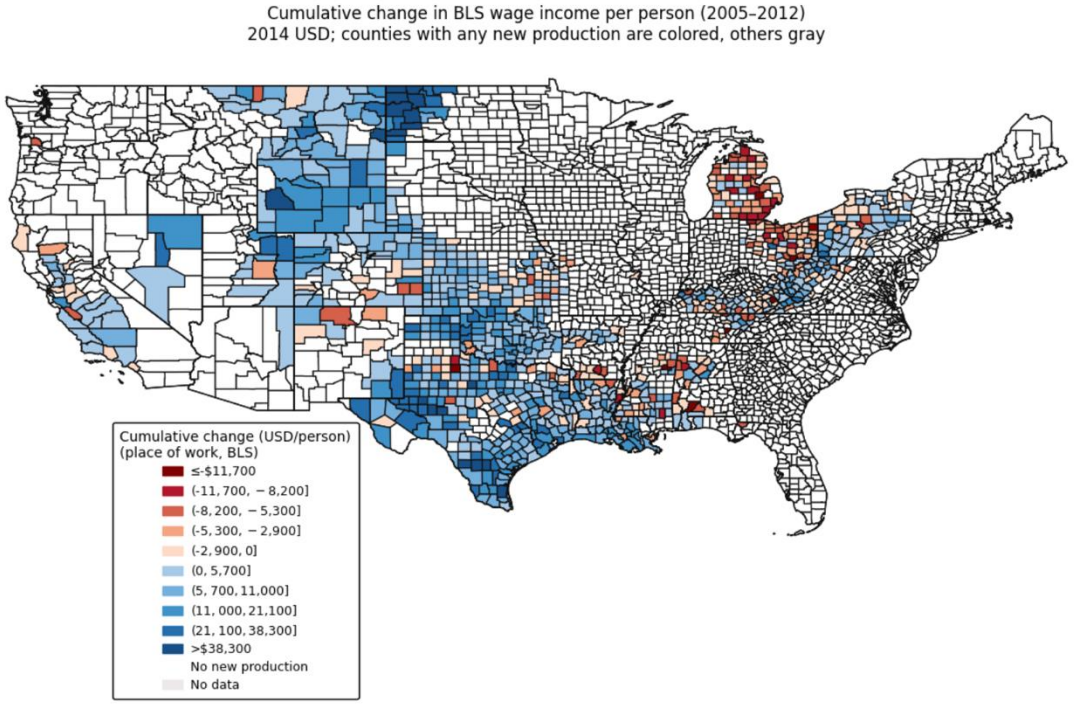

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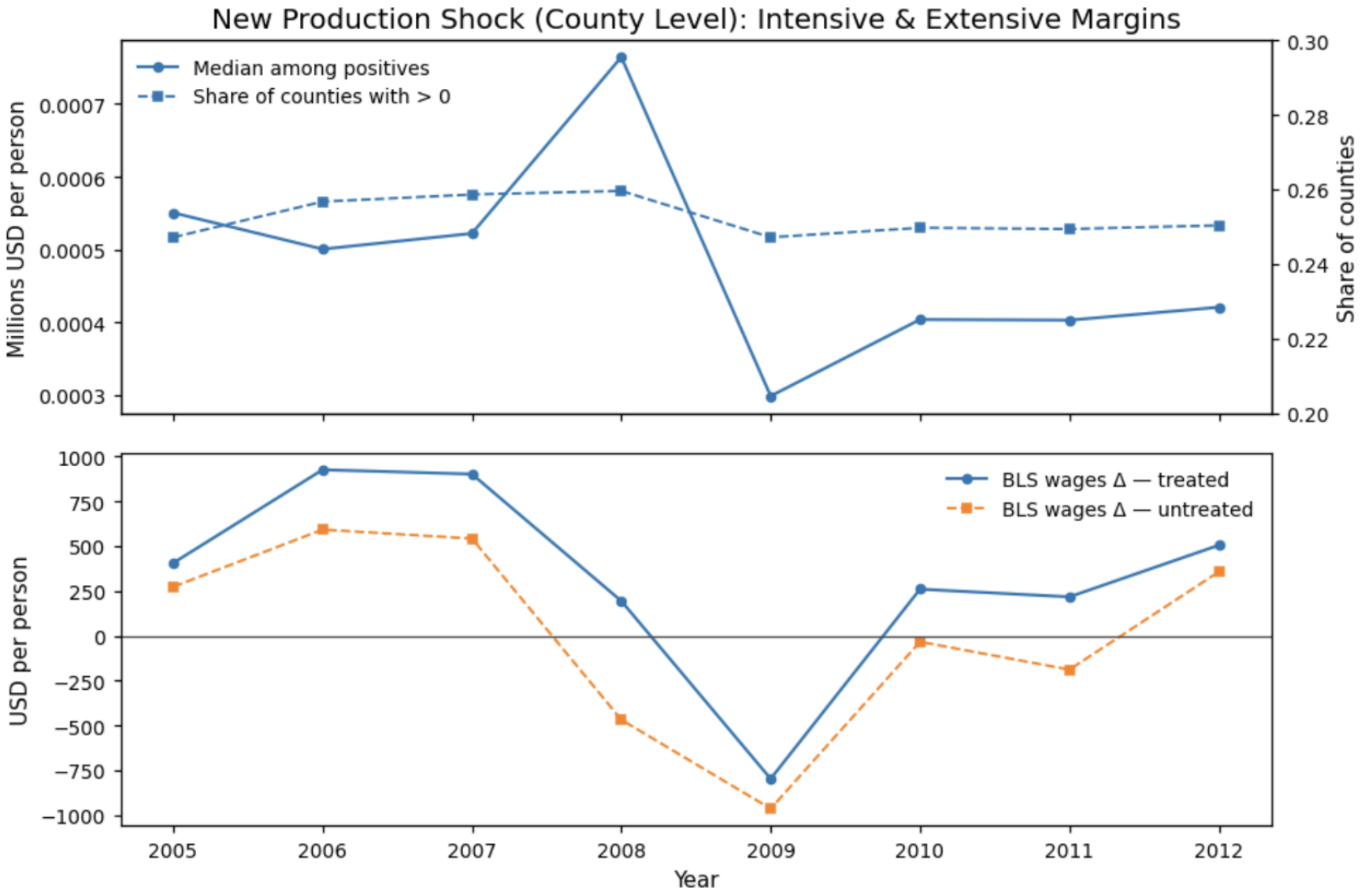

New Production Shock (County Level): Intensive & Extensive Margins
Median among positives
Share of counties with > 0
Millions USD per person
Share of counties
0.0007
0.0006
0.0005
0.0004
0.0003
0.30
0.28
0.26
0.24
0.22
0.20
BLS wages Δ — treated
BLS wages Δ — untreated
USD per person
1000
750
500
250
0
−250
−500
−750
−1000
2005
2006
2007
2008
2009
2010
2011
2012
Year

Econometrics

TWFE Model

$$\Delta Y_{it} = \beta_1 \times NewValue_{it} + \beta_2 \times NewValue_{i,t-1} + a_i + \omega_t + \epsilon_{i,t}$$

Variable	Meaning
ΔY_{it}	One year change in BLS Wages ($/person) , One year change in IRS other income, (defined as IRS AGI minus IRS wages) ($/person)
$NewValue_{it}$	Value of new production ($millions/person)
$NewValue_{i,t-1}$	Lagged $NewValue_{it}$
a_i	County FE
ω_t	Year FE

Two-way clustered SEs by county and year

OLS Results

$$\Delta Y_{it} = \beta_1 \times NewValue_{it} + \beta_2 \times NewValue_{i,t-1} + a_i + \omega_t + \epsilon_{i,t}$$

- Sample: 21,546 county-years (3078 counties x 7 years)
- All estimates from the replication matches that of the paper's
- Combining BLS wages and IRS other income lets us see how much of the production value remains locally
- **Interpretation:** Each coefficient is dollars of outcome per $1 million of new production

ΔY_{it} = Δ BLS wages	Estimate
β_1	33,957 (10,430)
β_2 (Not S.S.)	4,717 (11,339)

ΔY_{it} = Δ IRS other inc.	Estimate
β_1	64,094 (7,710)
β_2	-43,873 (5,781)

Endogeneity

- New production depends on both the availability of the resource and **firms deciding to extract it**.
- **Selection** may not be **exogeneous**. Firms choose *where* and *when* to drill based on local conditions like cheaper inputs, better infrastructure, favorable local cycles, etc.
  - These time-varying, county specific factors also comove with drilling and moves income even after FE.
- Reverse causality: wages → drilling
- We need an IV to isolate geological timing shocks so that surges in drilling opportunities are caused by geology + technology, not by local economies.

IV Intuition

- Idea: Predict each county's new production from **the timing of its' shale play's surge** taking off (shift) scaled by the county's **fixed exposure** to that play (share)
- "Predicted new production = Play timing x Play exposure"

"How much would County i produce in year t if it simply moved with its play's wave, scaled by County i's exposure?"

IV Mechanism (Bartik)

$$(1)\ \ln(NewValue_{it} + 1) = a_i + \lambda_{jt} + \epsilon_{i,t}$$

$$(2)\ \widehat{NewValue}_{it} = (e^{\widehat{a_i} + \widehat{\lambda_{jt}}} - 1)/pop_{it}$$

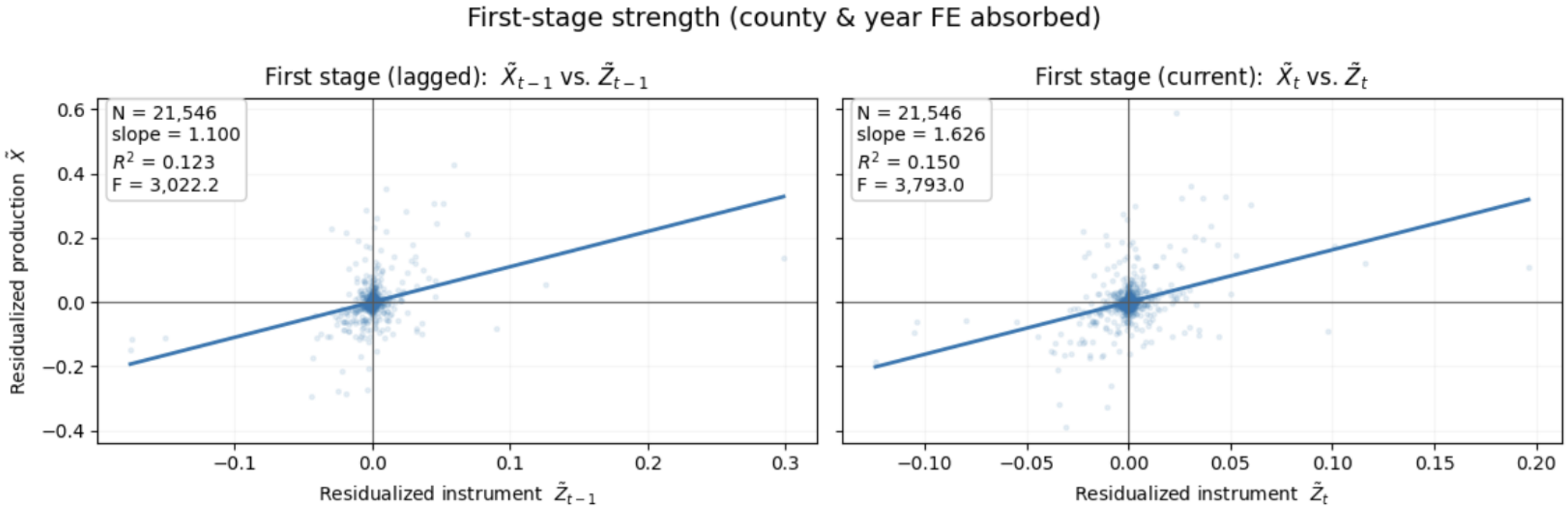
Variable	Meaning
a_i	Dummy for each county (captures share/exposure)
λ_{jt}	Set of dummies for each play-year combination (captures the wave/shift)

"The predicted values for new production are based on the timing of new production for all the counties within a particular play"

(Extension)

Informal First Stage Evidence

(Residualized variables, OLS, no clustering, one instrument at a time)



(Extension)

Formal First Stage Evidence

$$\widetilde{NewValue}_{i,t-1} = \pi_1 \times \tilde{Z}_{it} + \pi_2 \times \tilde{Z}_{i,t-1} + a_i + \omega_t + \epsilon_{i,t}$$

First stage (lagged)	Estimate
π_1	-0.543
$\boldsymbol{\pi_2}$	**1.441**

Joint F: 18.4 (p=0.0001)
R^2 = 0.13

$$\widetilde{NewValue}_{it} = \pi_1 \times \tilde{Z}_{it} + \pi_2 \times \tilde{Z}_{i,t-1} + a_i + \omega_t + \epsilon_{i,t}$$

First stage (current)	Estimate
$\boldsymbol{\pi_1}$	**1.567**
π_2 (Not S.S.)	0.068

Joint F: 10.1 (p=0.0065)
R^2 = 0.15

Relevant: the instrument and its lagged has strong predictive power for the endogenous regressor and its lag (after FE)

IV / 2SLS Results

$$\Delta Y_{it} = \beta_1 \times NewValue_{it} + \beta_2 \times NewValue_{i,t-1} + a_i + \omega_t + \epsilon_{i,t}$$

- Each million dollars of new production produces $80,000 in wage income and $132,000 in royalty and business income (IRS other income) within a county
  - 20% of production value remains in the county

ΔY_{it} = Δ BLS Wages	Estimate
β_1	79,751 (9,394)
β_2 (Not S.S.)	-12,534 (7,505)

ΔY_{it} = Δ IRS other inc.	Estimate
β_1	132,016 (23,358)
β_2	-113,855 (32,988)

Local spillovers model

$$\Delta Y_{it} = \beta_1 \times NewValue_{it} + \beta_2 \times NewValue_{i,t-1} + a_i + \omega_t + \epsilon_{i,t}$$

- To capture local spillovers, we built a distance panel by aggregating variables (and IVs) over all counties j whose centroid is within X miles of county i (inclusive)
  - Preferred specification is 100 miles

Variable	New meaning
ΔY_{it}	One year changes in BLS wages or IRS other inc., aggregated over X miles of county i ($/person)
$NewValue_{it}$	Sum of value of new production across all counties within X miles of county i ($millions/person)
$NewValue_{i,t-1}$	Lagged $NewValue_{it}$
a_i	Same (focal county) FE
ω_t	Same year FE

FE-IV (2SLS)

$$\Delta Y_{it} = \beta_1 \times NewValue_{it} + \beta_2 \times NewValue_{i,t-1} + a_i + \omega_t + \epsilon_{i,t}$$

- Within 100 miles, $1M of new production generates $257K in wage income and $286k in royalty and business income (IRS other income)
  - 54% of production value remains within 100 miles

ΔY_{it} = Δ BLS Wages	Estimate
β_1	257,252 (20,455)
β_2 (Not S.S.)	-26,821 (17,045)

ΔY_{it} = Δ IRS other inc.	Estimate
β_1	286,000 (30,632)
β_2	-306.864 (52,235)

By Industry (j)

$$\Delta Y_{itj} = \beta_{1j} \times NewValue_{it} + \beta_{2j} \times NewValue_{i,t-1} + a_i + \omega_t + \epsilon_{i,t,j}$$

- Using the preferred 100 mile IV, BLS wages specification
  - Mining accounts for roughly 60% of immediate income effect
  - Immediate effects are sectorally concentrated but later spill over…

Total	Mining	Trans.	Const.	Manuf.	Ed./Health	Govt.	Oth. Serv.
257,252 (20,455)	155,886 (14,546)	61,374 (8,073)	36,201 (3,413)	-16,006 (3,256)	1,433 (1,651)	10,346 (6,822)	8,018 (3,471)

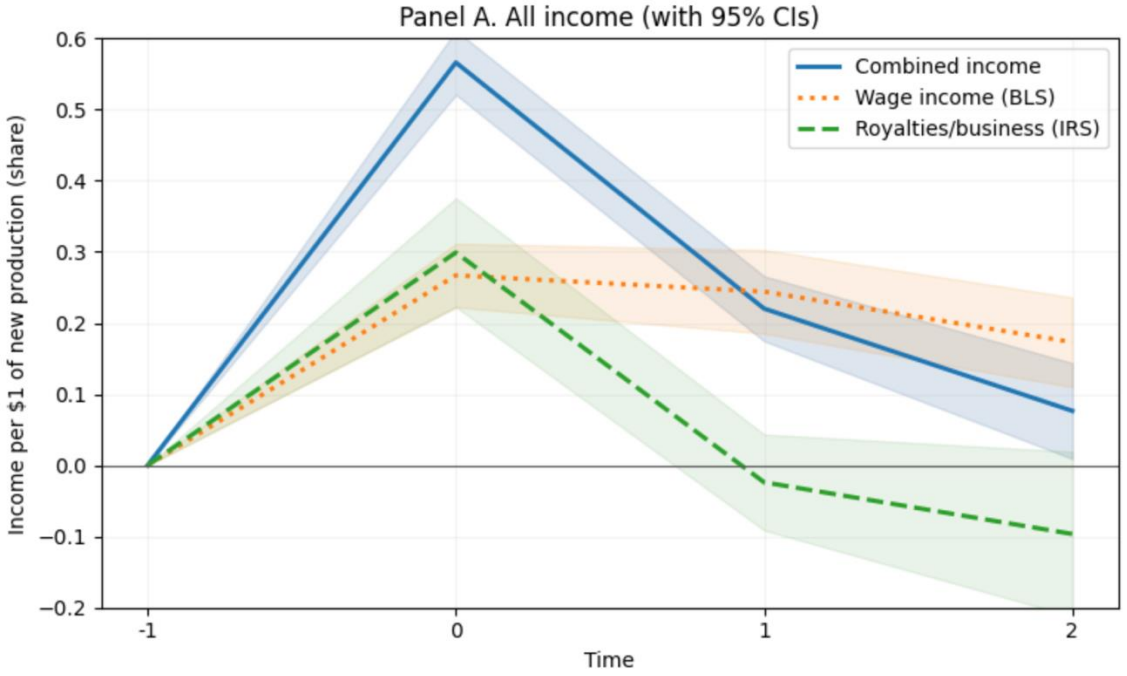
Dispersion of income across space and time

Income Effect Across Space

Income Effect Over Time (100-mile)

- Roughly two-thirds of the wage income increase persists for two years; royalties are short lived
- Composition of wage income increase shifts from extraction to spillover sectors

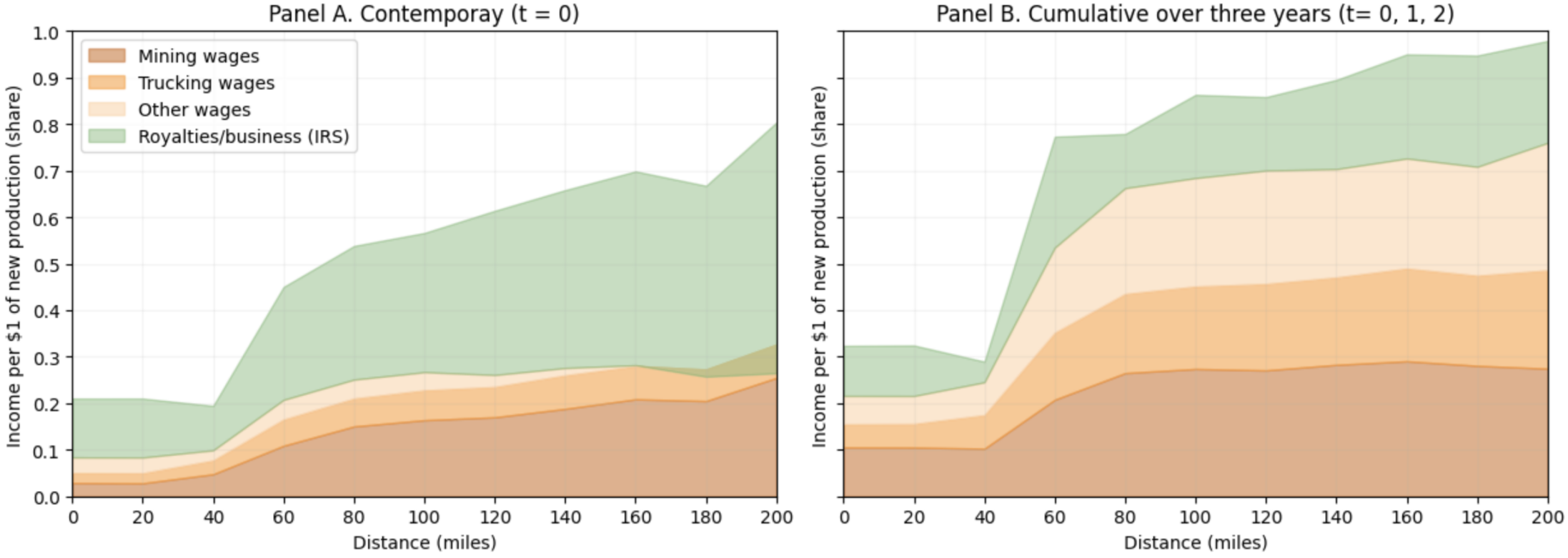
Figure 4. BLS and IRS Income Increases from New Production over Time



Effects Over Space and Time

- In the county alone, the 3-year cumulative effect equals about 33% of the revenue from the initial production. The effect exceeds 80% of the revenue in a 100-mile radius.

Figure 5. Income Effects over Time and Distance



Conclusion

- Local county effects are large but capture only part of the boom
  - 20% of production value remain in the county in the year of the shock
- Most of the value is captured regionally within 100 miles of the focal county
  - 54% of production value stays within 100 miles in the year of the shock
- Effects are persistent for wages, but short-lived for royalties
- Boom is initially concentrated in extraction, but spills across industries
- Over three years, 33% of initial production value stays in the county and 80% within 100 miles of the county

Policy Takeaways

- **Don’t overpromise locally:** Only a small share of value stays in the drilling county → be cautious with long-run spending commitments.
- **Plan regionally, not just locally:** Most of the gains show up within ~100 miles → coordinate transportation, housing, and infrastructure across counties.
- **Align costs and benefits:** Drilling counties bear the impacts; neighbors share the income → consider impact fees and regional revenue-sharing.
- **Look beyond extraction:** Mining jobs fade; transport and services persist → use boom years to invest in broader skills and infrastructure to avoid boom-bust.